

ATPL EASA STUDY GUIDE | 062 - RADIO NAVIGATION

Chapter 11: Performance-Based Navigation (PBN) & Specifications

Syllabus Source: CAE Oxford (pp. 425-472) | Pass Target: >= 90% AviationExam

1. THE PBN REVOLUTION: RNAV VS RNP

Performance-Based Navigation specifies aircraft Required Navigation Performance rather than mandating specific ground sensors:

- The Fundamental Distinction: RNP REQUIRES ON-BOARD PERFORMANCE MONITORING AND ALERTING (OPMA)! RNAV DOES NOT! Under RNP, the system continuously monitors containment and alerts the crew if Total System Error exceeds the containment boundary.
- Containment Standard: The aircraft navigation system must keep the aircraft within the designated accuracy value (e.g. +/-1.0 NM for RNP 1) for AT LEAST 95% OF TOTAL FLIGHT TIME, and within twice that value (2 x RNP) for 99.999% of time!

2. SUMMARY OF OFFICIAL PBN SPECIFICATIONS

PBN Specification	Accuracy Limit (95% Containment)	Flight Phase & Operational Application
PBN Specification	Accuracy Limit (95% Containment)	Flight Phase & Operational Application
RNAV 10 (RNP 10)	+/- 10 Nautical Miles	Oceanic and remote continental airspace (e.g. Pacific routes).
RNAV 5 (B-RNAV)	+/- 5 Nautical Miles	En-route continental airspace (Mandatory in European airspace).
RNAV 1 & RNAV 2 (P-RNAV)	+/- 1 NM or +/- 2 NM	Terminal arrivals (STAR) and departures (SID) with radar coverage.
RNP 4	+/- 4 Nautical Miles	Oceanic operations with reduced separation (30/30 NM).
RNP 1	+/- 1 Nautical Mile	Terminal arrivals (STAR) and departures (SID) without radar.
RNP APCH	+/- 0.3 NM on final approach	Non-precision approaches: LNAV, LNAV/VNAV, and LPV.
RNP AR APCH	Down to +/- 0.1 NM (Authorisation Required)	Complex terrain procedures with constant Radius-to-Fix (RF) curved legs!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): PBN Navigation Specifications R

PBN Accuracy Table (95% of total flight time): - RNAV 10 (RNP 10): Oceanic and remote areas (+/- 10 NM lateral accuracy). - RNAV 5 (B-RNAV): Continental en-route airspace in Europe (+/- 5 NM). - RNAV 1 / RNAV 2 (P-RNAV): Terminal arrivals and departures (+/- 1 NM). - RNP 1: SIDs and STARs in busy terminal airspace with mandatory on-board alerting. - RNP APCH: Standard RNAV (GNSS) approach to LNAV minima (+/- 0.3 NM in final segment). - RNP AR APCH: Authorization Required curved approaches down to RNP 0.1!

3. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: What is the fundamental operational difference between an 'RNAV' specification and an 'RNP' specification?

- [A]: RNP uses only satellites
- [B]: RNP REQUIRES ON-BOARD PERFORMANCE MONITORING AND ALERTING (OPMA), whereas RNAV does not
- [C]: RNAV is more accurate than RNP
- [D]: RNAV is for military only

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Under ICAO Doc 9613, the presence of real-time on-board performance monitoring and alerting (OPMA) is the sole distinguishing feature that defines an RNP specification.

Q2: In European en-route airspace, what is the mandatory minimum navigation specification known as Basic R?

- [A]: RNAV 1
- [B]: RNAV 5 (requiring aircraft to maintain track within +/- 5 NM for at least 95% of flight time)
- [C]: RNP 4
- [D]: RNAV 10

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. B-RNAV in Europe is formally designated as RNAV 5, mandating a lateral accuracy of 5 NM for 95% of the flight duration.

Q3: What unique procedure design capability is enabled by RNP AR APCH (Authorisation Required)?

- [A]: Vertical take-offs
- [B]: RADIUS-TO-FIX (RF) CURVED LEGS in the final approach and missed approach segments to skirt around high terrain
- [C]: Supersonic flight
- [D]: Zero fuel reserves

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. RNP AR allows precise curved RF turns in mountainous terrain with containment down to RNP 0.1.